

1.8 - Impedances

* The main definitions and convention choices of this section were based on [de Sá, PhD Thesis, 2018], while the remarks/properties also took support from the mentioned books of [A.W. Chao], [K.Y. Ng] and [S. Khan].

To further push the power/practicality of the wake formalism - which was up until this point entirely focused on the time/space-domain - we can look at wakes and their effect from a frequency-domain perspective. For this, we introduce the concept of impedances, which are proportional to the Fourier Transforms w.r.t. z of the wake functions:

$$\begin{aligned} Z_{||}(\omega) &= \frac{1}{c} \int_{-\infty}^{\infty} W_{||}(z) e^{i\omega z/c} dz & \text{(I)} \\ \vec{Z}_{\perp}(\omega) &= -\frac{i}{c} \int_{-\infty}^{\infty} \vec{W}_{\perp}(z) e^{i\omega z/c} dz & \text{(II)} \end{aligned}$$

$\hookrightarrow (Z_x, Z_y)$

* In most book ref's you'll find scalar $Z_{\perp}$ and $W_{\perp}$, which is due to considering a cyl. chan
 $\sim \vec{W}_{\perp} = W_{\perp} \cdot \hat{\rho}$

Going back to the more gen. transv. Taylor expansion of $W(\vec{\rho}_s, \vec{\rho}_w, z)$, from which the main wake

terms were $W_0'(z)$ for $W_{||}(z)$, and $W_x^D(z)$, $W_y^D(z)$ and $W^Q(z)$ for $\vec{W}_{\perp}(\vec{\rho}_s, \vec{\rho}_w, z)$, we get

the main impedance terms - $Z_{||}(\omega)$, $Z_x^D(\omega)$, $Z_y^D(\omega)$, $Z^Q(\omega)$ - expressions by making $W_{||}(z) \rightarrow W_0'(z)$ in (I) and $\vec{W}_{\perp}(z) \rightarrow W_x^D(z)$, $W_y^D(z)$ or $W^Q(z)$ in (II), respectively.

Alternatively, we could've defined one generalized impedance:

$$Z(\vec{\rho}_s, \vec{\rho}_w, \omega) = -\frac{i}{c} \int_{-\infty}^{\infty} W(\vec{\rho}_s, \vec{\rho}_w, z) e^{i\omega z/c} dz,$$

and then, expanding $W(\vec{\rho}_s, \vec{\rho}_w, z)$ like before, find all specific impedance terms:

$$Z(\vec{\rho}_s, \vec{\rho}_w, \omega) = \frac{c}{\omega} Z_{||}(\omega) + \overset{\textcircled{\#}}{x_w} \left[\overbrace{Z_x^D(\omega) x_s}^{Z_x} - \frac{1}{2} Z^Q x_w \right] + y_w \left[\overbrace{Z_y^D(\omega) y_s}^{Z_y} + \frac{1}{2} Z^Q y_w \right]$$

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$$\frac{c}{\omega} Z_{||} = \frac{c}{\omega} \cdot \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial W_0}{\partial z} e^{i\omega z/c} dz = \frac{1}{\omega} \left[W_0 e^{i\omega z/c} \Big|_{-\infty}^{\infty} - \frac{i\omega}{c} \int_{-\infty}^{\infty} W_0 e^{i\omega z/c} dz \right] = -\frac{i}{c} \int_{-\infty}^{\infty} W_0 e^{i\omega z/c} dz \quad \checkmark$$

Some important remarks:

1. The impedances $Z_{||}$, $\vec{Z}_{\perp}$ have unit of Ohms (Ω) - $\frac{1}{c} \cdot \omega \cdot dz \sim (\text{s/m}) \cdot (\text{V/c}) \cdot \text{m} = \frac{\text{V}}{\text{c/s}} = \frac{\text{V}}{\text{A}} = \Omega$ - , analogously to the concept of impedance in electrical engineering. ($Z_{x/y}^0$, Z^0 are in Ω/m)

2. The sign of the exponent in $e^{i\omega z/c}$ is a convention choice, and is equiv. to choosing to write the harmonic component of frequency ω of the beam current (at azimuth s and time t) as

$$I(s, t) = \hat{I} e^{-i\omega(t-s/c)}, \text{ together with } z > 0 \text{ behind the source particle.}$$

To understand this, let's look at how one can naturally get to $Z(\omega)$ when starting from $I(s, t)$:

*Adapted from [K.Y. Ng]

Consider a beam with such current component, $I(s, t) = \hat{I}(\omega) e^{-i\omega(t-s/c)}$. A particle at s will experience

the wake fields from all particles ahead of it, thus with positions $s+z$, and with charges

$$dq(z) = I(s+z, t) \cdot dt = \hat{I} e^{-i\omega(t-s/c-z/c)} \cdot dt = I(s, t) \cdot e^{i\omega z/c} \cdot \frac{dz}{c}.$$

(sometimes normalized by the total charge)

will experience a total voltage - or effective wake potential - given by the sum of all

$W_{||}(z) dq(z)$ terms, or better, by the integral of $W_{||}(z)$ times the source charge slices for all

$z > 0$, i.e.:

$$V(s, t) = \int_0^{\infty} W_{||}(z) \cdot \frac{I(s+z, t)}{c} dz = I(s, t) \cdot \frac{1}{c} \int_0^{\infty} W_{||}(z) \cdot e^{i\omega z/c} dz$$

$0 \rightarrow \text{can be set to } -\infty \text{ since } W(z < 0) = 0$

Then, since V is basically a convolution of I and $W_{||}$, it follows that we can also write $V(s, t) = \hat{V}(\omega) e^{-i\omega(t-s/c)}$, and thus:

$$\hat{V}(\omega) e^{-i\omega(t-s/c)} = \hat{I}(\omega) e^{-i\omega(t-s/c)} \cdot \frac{1}{c} \int_{-\infty}^{\infty} W_{||}(z) e^{i\omega z/c} dz,$$

and we get a very natural definition of the long. imp.:

$$Z_{||}(\omega) = \frac{\hat{V}(\omega)}{\hat{I}(\omega)} = \frac{1}{c} \int_{-\infty}^{\infty} W_{||}(z) \cdot e^{+i\omega z/c} dz$$

* Adapted from [S. Khan]

Another - more general - way of arriving at such definition of the impedance is starting with the effective "voltage" felt by a particle due to the wakes of an extended charge distribution, given by the convolution of $w_{||}(z)$ with the charge distribution $\lambda(z)$:

$$V_{||}(z) = \int_{-\infty}^{\infty} w_{||}(z') \cdot \lambda(z-z') dz' = \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z') \cdot I(z-z') dz',$$

$\lambda \sim C/m = c/v_m \cdot q/m$
 $\Rightarrow \lambda = I/c$

~ now adopting the synchronous position as the origin of the long. coord., so everything can be written in terms of z

and then expressing $V_{||}(z)$ and $I(z)$ in terms of their Fourier Transforms:

$$V_{||}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{V}_{||}(\omega) \cdot e^{-i\omega z/c} d\omega; \quad I(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{I}(\omega) e^{-i\omega z/c} d\omega$$

$\rightarrow \mathcal{F}^{-1}\{\hat{V}_{||}(\omega)\}$

Convention choice:

- $F(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt$
- $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{-i\omega t} d\omega$

implies:

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{V}_{||}(\omega) e^{-i\omega z/c} d\omega &= \frac{1}{2\pi} \cdot \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z') \cdot \int_{-\infty}^{\infty} \hat{I}(\omega) e^{-i\omega(z/c - z'/c)} d\omega dz' = \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{I}(\omega) e^{-i\omega z/c} \cdot \left(\frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z') e^{+i\omega z'/c} dz' \right) d\omega \end{aligned}$$

Then, defining $Z_{||}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z) e^{i\omega z/c} dz$, we get: $\hat{V}_{||}(\omega) = Z_{||}(\omega) \hat{I}(\omega)$, as intended!

* This latter expression is highly important for real-world applications because, in practice, num. simulations enable directly calculating the effective potentials $V(z)$, but not the wake functions $w(z)$ themselves, which are defined for point charges. Then, with the knowledge of $V(z)$ and $I(z)$, one can go to the freq. domain, calculate $Z(\omega) = \frac{\hat{V}(\omega)}{\hat{I}(\omega)}$, and then recover the actual $w(z)$'s through the inverse transform.

3. The additional $-i$ multiplicative factor in the transverse impedance terms is also an usual convention choice, and carries the purpose of making them look more like an ordinary impedance - where the real part is related to the dissipative/growth-damping effects.

↳ Longitudinal wake $\rightarrow$ acts on the dir. of motion $\rightarrow$ naturally performs work

$\rightarrow$ resulting directly in an energy dissipation $\sim \langle P \rangle \propto |\hat{I}|^2 \operatorname{Re}\{Z_{||}(\omega)\}$

↳ Transverse wake $\rightarrow$ acts perp. to the dir. of motion $\rightarrow$ generates a transverse kick $\Delta p \sim \Delta \theta$

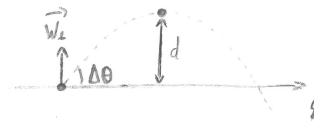
$\rightarrow$ such kick will be translated into transverse displacement only after a $\pi/2$ phase advance (betatron motion) $\rightarrow$ a factor $\pm i$ is needed to put the "dissipative" part of the wake in its real component

$\Rightarrow$ With such convention we assure that both $Z_{||}$

and $Z_{\perp}$ align with the standard physical meaning

of impedances, with $\operatorname{Re}\{Z\}$ being the part associated with

irreversible exchanges - net long. energy loss for $Z_{||}$, growth or damping of coherent transverse oscillations for $Z_{\perp}$ - while $\operatorname{Im}\{Z\}$ is the "reactive" part - mainly energy redistribution ($Z_{||}$) or coherent tune shifts ($Z_{\perp}$).



Furthermore, such convention leads to a "cleaner" freq-domain version of the Panofsky - Wenzel

Theorem: $\nearrow$ considers more terms of $\frac{\partial}{\partial z} W$ (not just W_0')

$$Z_{||}^{\text{full}}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z) e^{i\omega z/c} dz \quad \Rightarrow \quad \vec{\nabla}_{\perp} Z_{||}^{\text{full}}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} \vec{\nabla}_{\perp} w_{||}(z) e^{i\omega z/c} dz \stackrel{\text{P.W.}}{=} \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial \vec{w}_{\perp}}{\partial z} e^{i\omega z/c} dz$$

$$\Rightarrow \vec{\nabla}_{\perp} Z_{||}^{\text{full}}(\omega) = \frac{1}{c} \left(\left[\vec{w}_{\perp}(z) e^{i\omega z/c} \right]_{z=-\infty}^{z=\infty} - \frac{i\omega}{c} \int_{-\infty}^{\infty} \vec{w}_{\perp}(z) e^{i\omega z/c} dz \right) = \frac{\omega}{c} \vec{Z}_{\perp}(\omega) \quad \therefore \boxed{\vec{Z}_{\perp} = \frac{c}{\omega} \vec{\nabla}_{\perp} Z_{||}^{\text{full}}}$$

$\nearrow$ fields disappear at $\pm \infty$

$$= (W_x^D x_s - W_x^Q x_w, W_y^D y_s + W_y^Q y_w)$$

REVIEW + CLARIFICATIONS

Wakes:

- $W \approx W_0 + x_w \left(W_x^D x_s - \frac{1}{2} W^Q x_w \right) + y_w \left(W_y^D y_s + W^Q y_w \right)$
- $W_{||} \triangleq \frac{\partial}{\partial z} W(\vec{\rho}_s, \vec{\rho}_w, z) \approx W'_0(z) \sim \text{consideration only for the long. problem}$
- $\vec{W}_\perp \triangleq \vec{\nabla}_\perp W = \left(\frac{\partial}{\partial x_w} W, \frac{\partial}{\partial y_w} W \right) = (W_x, W_y)$
 $\Rightarrow \vec{W}_\perp \approx \left(W_x^D x_s - W^Q x_w, W_y^D y_s + W^Q y_w \right)$
- P.W. $\rightarrow \frac{\partial}{\partial z} \vec{W}_\perp = \vec{\nabla}_\perp W_{||}$ $\rightarrow$ here, $W_{||} \approx W'_0(z)$ shouldn't be used, otherwise $\vec{W}_\perp = 0$ always
- Cylind. symmetry: $W_x^D = W_y^D = W^D, W^Q = 0$
 $\Rightarrow \vec{W}_\perp = W^D (x_s, y_s) = W^D \vec{\rho}_s = W_\perp \hat{\rho}_s$

Impedances:

- $Z \approx \frac{\omega}{c} Z_{||} + x_w \left(Z_x^D x_s - \frac{1}{2} Z^Q x_w \right) + y_w \left(Z_y^D y_s + \frac{1}{2} Z^Q y_w \right)$
- Where:
 $Z \triangleq -\frac{i}{c} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz$; $Z_{||} \triangleq \frac{1}{c} \int_{-\infty}^{\infty} W'_0 e^{i\omega z/c} dz$;
 $Z^{D/Q} \triangleq -\frac{i}{c} \int_{-\infty}^{\infty} W^{D/Q} e^{i\omega z/c} dz$
- $Z_{||}^{\text{full}} = \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial W}{\partial z} e^{i\omega z/c} dz \stackrel{(\#)}{=} \frac{\omega}{c} Z$
 $\Rightarrow Z_{||}^{\text{full}} = Z_{||} + \frac{\omega}{c} x_w \left(Z_x^D x_s - \dots \right) + \dots$
 $\sim Z_{||}^{\text{full}} \approx Z_{||}$

$$\frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial W}{\partial z} e^{i\omega z/c} dz = \frac{1}{c} W e^{i\omega z/c} \Big|_{z=-\infty}^{z=\infty} - \frac{i\omega}{c^2} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz = \quad (\#)$$

$$\Rightarrow Z_{||}^{\text{full}} = \frac{\omega}{c} \left(-\frac{i}{c} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz \right) = \frac{\omega}{c} Z$$

$$\vec{Z}_\perp \triangleq \vec{\nabla}_\perp Z = \left(\frac{\partial}{\partial x_w} Z, \frac{\partial}{\partial y_w} Z \right) = (Z_x, Z_y)$$

$$\Rightarrow \vec{Z}_\perp \approx \left(Z_x^D x_s - Z^Q x_w, Z_y^D y_s + Z^Q y_w \right)$$

- P.W. $\rightarrow \vec{Z}_\perp = \vec{\nabla}_\perp Z = \frac{c}{\omega} \vec{\nabla}_\perp Z_{||}^{\text{full}}$ $\rightarrow$ again, the app. $Z_{||}^{\text{full}}$ should be used only when dealing w/ the long. effect
- Cylind. symmetry:

$$Z_x^D = Z_y^D = Z^D, Z^Q = 0$$

$$\Rightarrow \vec{Z}_\perp = Z^D (x_s, y_s) = Z^D \vec{\rho}_s = Z_\perp \hat{\rho}_s$$

• Extra: Multipole expansion approach

$$w_m(r, \theta, z) = -\frac{1}{q} \text{Im} W_m(z) r^m \cos(m\theta), \quad \begin{cases} W_{||,m} = \frac{\partial}{\partial z} W_m \\ W_{\perp,m} = \vec{\nabla}_\perp W_m \end{cases}$$

$$W'_m(z) \triangleq -\frac{\partial}{\partial z} W_m(z)$$

$$Z_{||,m} \triangleq \frac{1}{c} \int_{-\infty}^{\infty} W'_m(z) e^{i\omega z/c} dz = -\frac{\omega}{c} \left(-\frac{i}{c} \int_{-\infty}^{\infty} W_m(z) e^{i\omega z/c} dz \right) = -\frac{\omega}{c} Z_{\perp,m}$$

Complement on P.W. for impedances

As we've seen, P.W. is a natural "consequence" of writing $W_{||} = \frac{\partial}{\partial z} W$ and $\vec{W}_\perp = \vec{\nabla}_\perp \cdot W$. Thus,

it is also incorporated when using the generalized impedance $Z = -\frac{i}{c} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz$, together with the

definitions $Z_{||}^{\text{full}} = \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial}{\partial z} W e^{i\omega z/c} dz$ and $Z^{\text{D/Q}} = -\frac{i}{c} \int_{-\infty}^{\infty} W^{\text{D/Q}} e^{i\omega z/c} dz$, leading to:

$$Z_{||}^{\text{full}} \approx \frac{1}{c} \int_{-\infty}^{\infty} \left[W'_0 + X_\omega \left(W_X^{\text{D}'} X_s - \frac{1}{2} W_\omega^{\text{Q}'} X_\omega \right) + y_\omega(\dots) \right] e^{i\omega z/c} dz = Z_{||} + X_\omega \left(\frac{1}{c} \int_{-\infty}^{\infty} W_X^{\text{D}'} e^{i\omega z/c} dz \cdot X_s - \dots \right) + \dots$$

$$= Z_{||} + X_\omega \left(Z_{X,||}^{\text{D}} \cdot X_s - \frac{1}{2} Z_{||}^{\text{Q}} \cdot X_\omega \right) + \dots$$

but also:

And thus:

$$Z_{||}^{\text{full}} \approx \frac{\omega}{c} Z = Z_{||} + X_\omega \left(\frac{\omega}{c} Z_X^{\text{D}} X_s - \frac{1}{2} \frac{\omega}{c} Z_\omega^{\text{Q}} X_\omega \right) + \dots$$

$\vec{Z}_\perp = \vec{\nabla}_\perp Z = (Z_X^{\text{D}} X_s - Z_\omega^{\text{Q}} X_\omega, Z_Y^{\text{D}} Y_s + Z_\omega^{\text{Q}} Y_\omega)_\perp$

We could call these: $Z_{X,\perp}^{\text{D}}, Z_\perp^{\text{Q}}$

$$\begin{aligned} \bullet Z_{X/Y,\perp}^{\text{D}} &= \frac{c}{\omega} Z_{X/Y,||}^{\text{D}} \\ \bullet Z_\perp^{\text{Q}} &= \frac{c}{\omega} Z_{||}^{\text{Q}} \end{aligned}$$

4. The wakes can be obtained back from the impedances by the inverse transforms:

$$W'_0(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Z_{||}(\omega) e^{-i\omega z/c} d\omega; \quad W'_t(z) = \frac{i}{2\pi} \int_{-\infty}^{\infty} Z_t(\omega) e^{-i\omega z/c} d\omega,$$

where W_t refers to W_x^D, W_y^D or $W^q \sim$ same for Z_t .

5. Reality Properties:

Since the wakes are real quantities - $W_0'^* = W_0'$, $W_t'^* = W_t'$ - the impedances satisfy

$$Z_{||}^*(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} W'_0(z) e^{-i\omega z/c} dz = Z_{||}(-\omega) \Rightarrow \begin{cases} \operatorname{Re} Z_{||}(-\omega) = \operatorname{Re} Z_{||}(\omega) \sim \text{even} \\ \operatorname{Im} Z_{||}(-\omega) = -\operatorname{Im} Z_{||}(\omega) \sim \text{odd} \end{cases}$$

$$Z_t^*(\omega) = + \frac{i}{c} \int_{-\infty}^{\infty} W'_t(z) e^{-i\omega z/c} dz = -Z_t(-\omega) \Rightarrow \begin{cases} \operatorname{Re} Z_t(-\omega) = -\operatorname{Re} Z_t(\omega) \sim \text{odd} \\ \operatorname{Im} Z_t(-\omega) = \operatorname{Im} Z_t(\omega) \sim \text{even} \end{cases}$$

6. Passivity and Positivity

Using the harmonic decomposition of the beam current, the total energy lost by the beam to the chamber can be written as:

$$U_{\text{loss}} = \frac{1}{\pi} \int_0^{\infty} |\hat{I}(\omega)|^2 \operatorname{Re} \{ Z_{||}(\omega) \} d\omega, \quad \text{where clearly } |\hat{I}(\omega)|^2 \geq 0.$$

Thus, for a passive chamber: $U_{\text{loss}} \geq 0$ for any current distribution

Since we can choose $\hat{I}(\omega)$ concentrated around any specific frequency ω $\rightarrow \operatorname{Re} \{ Z_{||}(\omega) \} \geq 0 \quad \forall \quad \omega \geq 0$

And since $\operatorname{Re}\{Z_{||}(-\omega)\} = \operatorname{Re}\{Z_{||}(\omega)\}$, this means:

$$\operatorname{Re}\{Z_{||}(\omega)\} \geq 0 \quad \text{for all } \omega$$

Now, for the transverse impedances, something similar can be derived for the dipolar terms:

→ For a displaced beam - described by a dipole-current moment $D(\omega) \sim x_s \hat{I}(\omega)$ - the contribution of the dipole channel to the energy loss can be written:

$$U_D \propto \int_0^\infty |D(\omega)|^2 \cdot \operatorname{Re}\{Z_{||,D}(\omega)\} d\omega \quad ; \quad \text{where} \quad Z_{||,D} = \frac{1}{c} \int_{-\infty}^\infty W_{x/y}^{D'}(z) e^{i\omega z/c} dz$$

(remembering that $W_{||}^{full}(z) = \frac{\partial}{\partial z} W(z) = W'_o(z) + x_w W_x^{D'} + y_w W_y^{D'} + \dots$)

Integrating by parts:

$$\frac{1}{c} \int_{-\infty}^\infty W_{x/y}^{D'} e^{i\omega z/c} dz = \frac{\omega}{c} \left(-\frac{i}{c} \int_{-\infty}^\infty W_{x/y}^D e^{i\omega z/c} dz \right) = \frac{\omega}{c} Z_{x/y}^D$$

And thus, passivity implies:

$$U_D \geq 0 \quad \text{for any } D(\omega) \Rightarrow \operatorname{Re}\{Z_{||,D}(\omega)\} \geq 0 \Rightarrow \frac{\omega}{c} \operatorname{Re}\{Z_{x/y}^D\} \geq 0 \quad \forall \omega \geq 0$$

$$\therefore \text{Since } \operatorname{Re}\{Z_t(-\omega)\} = -\operatorname{Re}\{Z_t(\omega)\} : \quad \Rightarrow \operatorname{Re}\{Z_{x/y}^D\} \geq 0 \quad \forall \omega \geq 0$$

$$\begin{cases} \operatorname{Re}\{Z_{x/y}^D(\omega)\} \geq 0, & \omega \geq 0 \\ \operatorname{Re}\{Z_{x/y}^D(\omega)\} \leq 0, & \omega < 0 \end{cases}$$

* For the quadrupolar impedance, passivity does not imply $\operatorname{Re}\{Z^Q(\omega)\} \geq 0$ because, while

$$U_D \propto |D|^2 \frac{\omega}{c} \operatorname{Re}\{Z_{x/y}^D(\omega)\} \quad \left(\text{which} \Rightarrow \omega \cdot \operatorname{Re}\{Z^D\} \geq 0 \right), \quad U_Q \propto \operatorname{Re}\left\{ Z^Q \overset{\text{monopole moment}}{I_o} \overset{\text{quad. moment}}{Q_b^*} \right\} \cdot \frac{\omega}{c}, \quad \text{and } I_o \cdot Q_b^* \text{ can have any sign}$$

7. Causality and Kramers - Kronig

Due to the causality of the wakes - $W(z < 0) = 0$ -, the real and imaginary parts of the impedances cannot be completely arbitrary: they satisfy the so-called Kramers - Kronig relations, i.e., are connected by a Hilbert transform.

To derive and outline such connexion, let's write $Z_{||}(\omega) = R_{||}(\omega) + iX_{||}(\omega)$.

Then, since the wakes are real:

$$\begin{cases} R_{||}(\omega) = \text{Re}\{Z_{||}(\omega)\} \Rightarrow R_{||}(-\omega) = R_{||}(\omega) \\ X_{||}(\omega) = \text{Im}\{Z_{||}(\omega)\} \Rightarrow X_{||}(-\omega) = -X_{||}(\omega) \end{cases}$$

From the inverse transform:

$$W_{||}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Z_{||}(\omega) e^{-i\omega z/c} d\omega = \frac{1}{2\pi} \left[\int_{-\infty}^0 Z_{||}(\omega) e^{-i\omega z/c} d\omega + \int_0^{\infty} Z_{||}(\omega) e^{-i\omega z/c} d\omega \right] =$$

$$= \frac{1}{2\pi} \left[\int_0^{\infty} (R_{||}(-\omega) + iX_{||}(-\omega)) e^{-i\omega z/c} d\omega + \int_0^{\infty} (R_{||}(\omega) + iX_{||}(\omega)) e^{-i\omega z/c} d\omega \right] =$$

$$= \frac{1}{2\pi} \int_0^{\infty} \left[R_{||}(\omega) \cdot \overbrace{(e^{i\omega z/c} + e^{-i\omega z/c})}^{2 \cos(\omega z/c)} + iX_{||}(\omega) \cdot \overbrace{(e^{i\omega z/c} - e^{-i\omega z/c})}^{-2i \sin(\omega z/c)} \right] d\omega =$$

$$= \frac{1}{\pi} \int_0^{\infty} [R_{||}(\omega) \cos(\omega z/c) + X_{||}(\omega) \sin(\omega z/c)] d\omega$$

Now, to make use of causality, let $u > 0$ and evaluate $w_{II}(z = -u)$:

$$w_{II}(-u) = \frac{1}{\pi} \int_0^{\infty} \left[R_{II}(\omega) \cos\left(\frac{\omega u}{c}\right) - X_{II}(\omega) \sin\left(\frac{\omega u}{c}\right) \right] d\omega = 0$$

$$\Rightarrow \int_0^{\infty} R_{II}(\omega) \cos\left(\frac{\omega u}{c}\right) d\omega = \int_0^{\infty} X_{II} \sin\left(\frac{\omega u}{c}\right) d\omega$$

Then, for $z > 0$, we can write (from (I)):

$$w_{II}(z) = \frac{2}{\pi} \int_0^{\infty} R_{II}(\omega) \cos\left(\frac{\omega z}{c}\right) d\omega \quad (\text{II}) \quad \text{or} \quad w(z) = \frac{2}{\pi} \int_0^{\infty} X_{II}(\omega) \sin\left(\frac{\omega z}{c}\right) d\omega \quad (\text{III})$$

$\Rightarrow *$ The knowledge of either $\text{Re}\{Z_{II}\}$ or $\text{Im}\{Z_{II}\}$ alone is enough to reconstruct the wake w_{II}

Now, to derive Kramers-Kronig relations we see that:

$$Z_{II}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} w_{II}(z) e^{i\omega z/c} dz = \frac{1}{c} \int_0^{\infty} \left[w_{II} \cos\left(\frac{\omega z}{c}\right) + i w_{II} \sin\left(\frac{\omega z}{c}\right) \right] dz = R_{II}(\omega) + i X_{II}(\omega)$$

$$\Rightarrow R_{II}(\omega) = \frac{1}{c} \int_0^{\infty} w_{II}(z) \cos\left(\frac{\omega z}{c}\right) dz \quad (\text{III}) = \frac{2}{\pi c} \int_0^{\infty} \int_0^{\infty} X_{II}(\omega') \sin\left(\frac{\omega' z}{c}\right) \cos\left(\frac{\omega z}{c}\right) d\omega' dz =$$

$$= \frac{2}{\pi c} \int_0^{\infty} \left(\int_0^{\infty} \frac{1}{2} \left[\sin\left(\frac{(\omega' + \omega)z}{c}\right) + \sin\left(\frac{(\omega' - \omega)z}{c}\right) \right] dz \right) d\omega'$$

$$\begin{aligned} \sin(a+b) &= \sin a \cos b + \sin b \cos a \\ \sin(a-b) &= \sin a \cos b - \sin b \cos a \\ \Rightarrow 2 \sin a \cos b &= \sin(a+b) + \sin(a-b) \end{aligned}$$

Using the strategy of introducing a small damping factor $e^{-\eta z}$ ($\eta > 0$)

$$\int_0^{\infty} \sin\left(\frac{az}{c}\right) dz = \lim_{\eta \rightarrow 0^+} \int_0^{\infty} e^{-\eta z} \sin\left(\frac{az}{c}\right) dz = \lim_{\eta \rightarrow 0^+} \frac{ac}{a^2 + \eta^2 c^2} = \frac{c}{a}$$

However, if "a" can pass through zero inside another, like in our case of $a = \omega' - \omega$, this result must be interpreted in the ^{Cauchy} principal value sense: $c \cdot P \frac{1}{a}$.

$$\frac{1}{\omega' + \omega} + \frac{1}{\omega' - \omega} = \frac{2\omega'}{\omega'^2 - \omega^2}$$

Thus:

$$R_{11}(\omega) = \frac{1}{\pi c} \int_0^{\infty} \left(\frac{c}{\omega' + \omega} + P \frac{c}{\omega' - \omega} \right) X_{11}(\omega') d\omega' = \frac{2}{\pi} P \int_0^{\infty} \frac{\omega' X_{11}(\omega')}{\omega'^2 - \omega^2} d\omega'$$

And analogously:

$$X_{11}(\omega) = \frac{1}{c} \int_0^{\infty} \omega_{11}(z) \sin\left(\frac{\omega z}{c}\right) dz \quad \xrightarrow{[II] + algebra} \quad X_{11}(\omega) = -\frac{2\omega}{\pi} P \int_0^{\infty} \frac{R_{11}(\omega')}{\omega'^2 - \omega^2} d\omega'$$

Or equivalently, opening $\frac{2\omega'}{\omega'^2 - \omega^2}$ back into $\frac{1}{\omega' + \omega} + \frac{1}{\omega' - \omega}$ and manipulating to express these relations in terms of the whole frequency axis, gives the standard Kramers - Kronig relations:

$$R_{11}(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{X_{11}(\omega')}{\omega' - \omega} d\omega'; \quad X_{11}(\omega) = -\frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{R_{11}(\omega')}{\omega' - \omega} d\omega'$$

Proceeding analogously for the transverse impedances ($Z_t(\omega) = R_t(\omega) + iX_t(\omega)$) also gives:

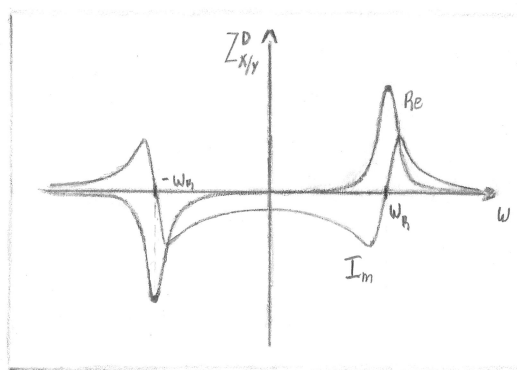
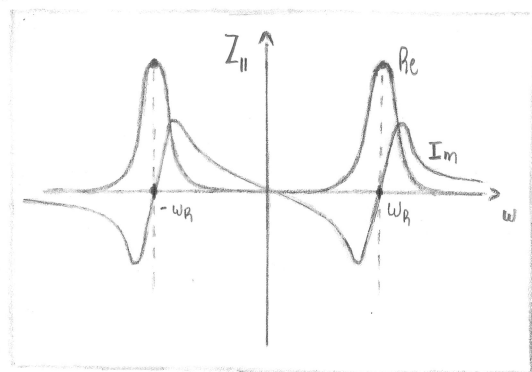
$$R_t(\omega) = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{X_t(\omega')}{\omega' - \omega} d\omega'$$

$$X_t(\omega) = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{R_t(\omega')}{\omega' - \omega} d\omega'$$

~ Kramers-Kronig relations for the transverse impedances

To illustrate many of the presented properties of the real and imaginary

parts of $Z_{||}(\omega)$ and $Z_t(\omega)$, we can draw these curves for the case of a resonator model - one of the models that will be explored in the next section - with resonant frequency ω_R :



Adapted from [S. Khan]